Close

```
**Experiment 8**
```

Experiment 8

Q1 Answer following question in typed form and paste it in the text box provided for submission: **bold text**

1 What is 2 tailed and tailed T Test ? **bold text**

t-Test Types: One-Tailed and Two-Tailed

A t-test is used in hypothesis testing to check whether there is a significant difference between means. Based on the hypothesis, it can be:

1. One-Tailed t-Test

Definition

A one-tailed t-test checks for a difference in only one direction (either greater than OR less than).

Types

(a) Right-Tailed Test Tests if value is greater than a specific value H 0 : μ

19 H 0

:μ=19 H 1 : μ

19 H 1

:μ>19 **(b) Left-Tailed Test** Tests if value is less than a specific value H 0 : μ

19 H 0

:μ=19 H 1 : μ < 19 H 1

:μ<19

Graph Idea Rejection region is only on one side of the distribution

When to Use When direction is already known or expected

2. Two-Tailed t-Test

Definition

A two-tailed t-test checks for a difference in both directions (greater than OR less than).

Hypothesis H 0 : μ

19 H 0

:μ=19 H 1 : μ ≠ 19 H 1

:μ □ =19

Graph Idea Rejection regions are on both sides (left & right tails)

When to Use When you want to check any difference, not specific direction

Key Differences

Feature	One-Tailed Test	Two-Tailed Test
Direction	One direction	Both directions

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Feature	One-Tailed Test	Two-Tailed Test
Alternative Hypothesis	(>) or (<)	(\neq)
Rejection Region	One side	Two sides
Significance Level	All α in one tail	α/2 in each tail

2 Explain what is Type I and Type II errors with an example. **bold text**

Type I and Type II Errors

In hypothesis testing, errors can occur when making decisions about the null hypothesis H 0 H 0

.

Definition

A Type I error occurs when we reject the null hypothesis H 0 H 0

even though it is actually true.

Probability Denoted by α (level of significance) Example: α = 0.05 → 5% chance of Type I error

Example

Suppose:

H 0 H 0

: A student's average age is 19

👉 Type I Error:

You conclude age is not 19, but actually it is 19

2. Type II Error (False Negative) Definition

A Type II error occurs when we fail to reject H 0 H 0

even though it is false. .

Probability Denoted by β

Example

Suppose:

H 0 H 0

: Average age is 19

** Type II Error:**

You conclude age is 19, but actually it is not 19

Difference Between Type I and Type II Errors

Error Type	Decision Taken	Reality	Meaning
Type I Error	Reject (H_0)	(H_0) is true	False alarm
Type II Error	Accept (H_0)	(H_0) is false	Missed detection

Real-Life Example (Medical Test) H 0 H 0

:** Patient is healthy** **Type I Error:** Test says patient is sick, but actually healthy **Type II Error:** Test says patient is healthy, but actually sick

3 What is power of statistical test?

Power of Statistical Test

Definition

The power of a statistical test is the probability of correctly rejecting the null hypothesis H 0 H 0

when it is actually false.

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Formula Power

1 − β Power=1−β

Where:

β β = probability of Type II error

Explanation *High power *→ Test is good at detecting true differences *Low power* → Test may miss real differences

Example

Suppose:

H 0 H 0

: Average age = 19

If the true average age is not 19 and the test correctly rejects H 0 H 0

, then the test has high power.

Factors Affecting Power **Sample Size (n)** Larger sample → higher power **Significance Level (α)** Higher α → higher power **Effect Size** Larger difference → higher power **Variability (Standard Deviation)** Less variation → higher power

Importance Helps in designing experiments Ensures reliable results Reduces chances of Type II error

4 Explain input and out put characteristics for stats.ttest_rel function in your own words

stats.ttest_rel() – Input and Output Characteristics

The stats.ttest_rel() function (from SciPy library) is used to perform a paired t-test.

It compares the means of two related samples (dependent samples).

Purpose To check whether there is a significant difference between two related measurements Example: Before vs After experiment Same students' marks in two tests

Input Characteristics

a = first sample b = second sample

Both must:

Have same length Be paired/dependent data

2. Data Type Input should be: Numerical values (list, array, or pandas series) ** 3. Assumptions** Data should be: Paired observations Approximately normally distributed differences

Output Characteristics

1. t-statistic (t_stat) Measures difference between paired samples Larger value → greater difference ** 2. p-value** Helps in decision making

Rule:

If p-value ≤ 0.05 → Reject H 0 H 0

If p-value > 0.05 → Fail to reject H 0 H 0

Example

```
from scipy import stats

before = [10, 12, 11, 14, 13]
after  = [12, 13, 12, 15, 14]

t_stat, p_value = stats.ttest_rel(before, after)

print("t-value:", t_stat)
print("p-value:", p_value)
```


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```
print("p-value:", p_value)
```

```
t-value: -5.999999999999998
p-value: 0.0038825370469605155
```

Interpretation If p-value is small → significant difference between before & after If p-value is large → no significant difference

5 What is function in stats module for identifying power of a statistical test. Explain out put range of it and its significance. **bold text**

Function to Identify Power of a Statistical Test

In Python, power of a statistical test is commonly calculated using the statsmodels module (not directly in basic scipy.stats).

Function Used

From statsmodels.stats.power:

Input Parameters effect_size → Difference between means (standardized) nobs → Number of observations (sample size) alpha → Significance level (e.g., 0.05)

Output Returns a single value: power of the test

Interpretation of Values

Power Value	Meaning
< 0.5	Weak test (poor detection)
≈ 0.8	Good power (standard value)
> 0.9	Very strong test

Common standard: Power ≥ 0.8

Significance of Power ** 1. Detects True Effect** Higher power → better chance of finding real difference ** 2. Reduces Type II Error** Since Power = 1 − β 1−β High power → low probability of Type II error **3. Helps in Experiment Design** Used to decide: Sample size Required effect size ** 4. Improves Reliability** Ensures conclusions are accurate and meaningful

```
from statsmodels.stats.power import TTestPower

analysis = TTestPower()
power = analysis.power(effect_size=0.5, nobs=30, alpha=0.05)

print("Power:", power)
```

```
Power: 0.7539647157437905
```

Conclusion The function TTestPower().power() is used to calculate power Output lies between 0 and 1 Higher power means better ability to detect true differences and more reliable statistical results

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